SVKM'S NMIMS
MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT& ENGINEERING
Academic Year: 2024-2025

Program: B Tech/MBA Tech/B.Tech (IT/Mech. – Intgr.) Year: I/II/IV

Stream: MBA Tech – (COMP/AI/IT)/ B.Tech (CE/CS/AI/ CS&E(CS)/DS/AIML/Mech./IT/AI&DS/Civil/IT(Intg.)/Mech(Intg.) Semester: II/IV/VIII

Time: 3 hrs. (10:00 am to 01:00pm)

Subject: Management Accounting for Engineers
Date: 19 /05/ 2025 No. of Pages: 04
Marks: 100 Marks

Final Examination/Re-examination

Instructions: Candidates should read carefully the instructions printed on the question paper and on the cover of the Answer Book, which is provided for their use.

1) Question No. _1___ is compulsory.
2) Out of the remaining questions, attempt any 4 questions.
3) In all ___5_ questions to be attempted.
4) All questions carry equal marks.
5) Answer of each new question to be started on a fresh page.
6) Figures in brackets on the right-hand side indicate full marks.
7) Assume Suitable data if necessary.

<table>
<tr><td>Q-1</td><td></td><td>Answer briefly:</td><td>20</td></tr>
<tr><td>CO- 1; SO- ;
BL- Remember</td><td>a</td><td>Who are the users of financial statements and what are their needs?</td><td>5</td></tr>
<tr><td>CO- 3 ; SO- ;
BL- Evaluate</td><td>b</td><td>Diwan Ltd. supplied the following information about a product :
• Selling price per unit: ₹50
• Direct material cost per unit: ₹15
• Direct labour cost per unit: ₹10
• Fixed overheads: ₹72,000
Variable overheads: 70% of direct labour cost
Compute:
1. The BEP in Units and Value
2. The profit if the total sales volume is 10,000 units</td><td>5</td></tr>
<tr><td>CO-2 ; SO- ;
BL- Evaluate</td><td>c</td><td>A company uses Product X with the following consumption data:
<table><tr><th>Usage Type</th><th>Quantity (units per week)</th></tr><tr><td>Normal usage</td><td>60 units</td></tr><tr><td>Minimum usage</td><td>30 units</td></tr><tr><td>Maximum usage</td><td>90 units</td></tr></table>Other inventory details:
Re-order quantity: 400 units
Re-order period: 3 to 5 weeks</td><td>5</td></tr>
</table>

		Calculate : (a) Re-order level (c) Maximum level (b) Minimum level (d) Average stock level	
CO- 4 ; SO- ; BL- Evaluate	d	From the following information calculate Material Cost Variance, Material Price Variance and Material Usage Variance. **Materials** | **Standard Quantity** | **Standard Rate (₹)** | **Actual Quantity** | **Actual Rate (₹)** X | 20 | 5 | 25 | 4.50 Y | 30 | 4 | 30 | 5.50	5
Q-2 CO-2; SO-; BL- Evaluate	a	The following are costing information related to ABC Company Ltd. for three months ending 31st March 2025. **Particulars** | **₹** Raw material (1st Jan 2025) | 5,500 Raw material (31st March 2025) | 4,000 Work in progress (1st Jan 2025) | 14,000 Work in progress (31st March 2025) | 15,000 Purchases of raw material | 30,000 Direct wages | 31,500 Rent and rates | 17,000 Carriage inwards | 2,000 Carriage outwards | 5,000 Supervisor's salary | 12,000 Administration cost | 31,000 Factory rent | 75,000 Royalty on manufacturing | 2,000 Warehouse rent | 10,000 Primary Packing material | 5,000 Repairs on machinery | 5,000 Indirect wages | 4,000 Travelling expense | 8,000 Managing Directors remuneration | 60,000 Depreciation on plant | 4,000 **Other information:** The selling price per unit is ₹9.5. The stock of finished goods as of 1st Jan 2025 is 2,000 units valued at ₹ 5 and on 31st March 2025 is 2,500 units. The company sold 39,000 units during the period. Prepare a cost sheet and calculate the total cost, profit and sales.	15
Q-2 CO-2; SO-; BL- Remember	b	Explain the meaning and purpose of ABC analysis of Inventory Management.	5

<table>
<tr><td>Q-3
CO- 3; SO-;
BL- Apply</td><td>a</td><td>The sales and profit for two years are as follows:<table><tr><td>Year</td><td>Sales (₹)</td><td>Profit (₹)</td></tr><tr><td>2020</td><td>2,40,000</td><td>35,000</td></tr><tr><td>2021</td><td>2,80,000</td><td>50,000</td></tr></table>Calculate:
(a) P/V Ratio (Profit-Volume Ratio)
(b) Break-even Sales (BEP) in ₹
(c) Revised BEP if fixed costs increase by 20%
(d) Expected profit if sales increase by 25% in the year 2022
(e) Margin of safety if profit of ₹2,00,000.</td><td>15</td></tr>
<tr><td>Q-3
CO- 3; SO-; BL-
Understand</td><td>b</td><td>Explain the concept of the Break-even Point (BEP). Why is it important for businesses?</td><td>5</td></tr>
<tr><td>Q-4
CO- 3; SO-;
BL- Apply</td><td>a</td><td>XYZ Ltd manufactures three products A, B, and C. The unit selling prices of these products are ₹120, ₹90, and ₹60, respectively. The corresponding unit variable costs are ₹70, ₹50, and ₹30. The proportions of A, B and C in which these products are manufactured and sold are 25%, 35%, and 40%, respectively. The total fixed costs are ₹19,25,000.

Requirements:
1. Calculate the overall break-even quantity and the product-wise break-up of break-even quantity.
2. Determine the total profit if the total sales quantity increases by 30% above the break-even quantity.
3. Find the margin of safety (in total) if actual sales (total) are ₹50,00,000.</td><td>15</td></tr>
<tr><td>Q-4
CO- 1; SO-;
BL- Remember</td><td>b</td><td>Explain the differences between Financial Accounting and Management Accounting.</td><td>5</td></tr>
<tr><td>Q-5
CO- 5; SO-;
BL-Apply</td><td>a</td><td>ABC Ltd. has furnished the following information relating to costs at a capacity level of 10,000 units.<table><tr><th>Particulars</th><th>₹</th></tr><tr><td>Stores</td><td>2,000 (100% variable)</td></tr><tr><td>Depreciation</td><td>20,000 (100% fixed)</td></tr><tr><td>Inspection</td><td>1,000 (20%variable)</td></tr><tr><td>Administration overheads</td><td>10,000 (25% variable)</td></tr><tr><td>Selling overheads</td><td>6,000 (50% variable)</td></tr><tr><td>Material cost</td><td>50,000 (100% variable)</td></tr><tr><td>Labour cost</td><td>30,000 (100% variable)</td></tr><tr><td>Repairs and maintenance</td><td>4.000 (75% variable)</td></tr><tr><td>Power</td><td>2,500 (80% variable)</td></tr></table>Prepare a production cost budget at the level of 12,000 units and 16,000 units.</td><td>5</td></tr>
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<table>
<tr><td>Q-5
CO- 5; SO-;
BL- Understand</td><td>b</td><td>What is Budgeting? Write the advantages of Budgeting.</td><td>5</td></tr>
<tr><td>Q-6
CO- 2, SO-;
BL- Evaluate</td><td>a</td><td>M/s Pratap Industries Ltd. are the manufacturers of Plastic Toys. The following data relate to the manufacturing of Toys during March 2023:
<table><tr><td>Raw Material Consumed</td><td>Rs. 20,000</td></tr><tr><td>Direct Wages</td><td>Rs. 12,000</td></tr><tr><td>Machine-hour worked</td><td>9,500 hours</td></tr><tr><td>Machine hour rate</td><td>Rs. 2 per hour</td></tr><tr><td>Office Overheads</td><td>20% of Works Cost</td></tr><tr><td>Selling Overheads</td><td>Re. 0.50 per unit</td></tr><tr><td>Units Produced</td><td>20,000</td></tr><tr><td>Units Sold</td><td>18,000 @ Rs. 5 per units</td></tr></table>You are required to prepare a Cost Sheet from the above information.</td><td>12</td></tr>
<tr><td>Q-6
CO-2; SO-;
BL- Remember
& Understand</td><td>b</td><td>ABC Ltd. purchases 1,600 units of a component each year at a unit cost of Rs. 15 per unit. The ordering cost is Rs. 100 per order and the holding cost per unit per year is computed at Rs. 8. You are required to calculate:
a) Economic Order Quantity(EOQ).
b) Total number of orders to be placed in a year as per EOQ
c) Time between two orders.
d) Total Cost of the Components.</td><td>8</td></tr>
<tr><td>Q-7
CO- 4; SO-;
BL- Evaluate</td><td>a</td><td>The following information is available from the records of ABC Ltd. for January 2025. The standard Output is 100 units. Standard hours taken and rate per hour were as follows:
<table><tr><th>Workers</th><th>Hours</th><th>Rate per Hour</th></tr><tr><td>Skilled labour</td><td>250</td><td>80</td></tr><tr><td>Semi – Skilled Labour</td><td>200</td><td>32</td></tr><tr><td>Unskilled Labour</td><td>400</td><td>48</td></tr></table>The Actual Production was 750 units, and the actual hours worked and rate per hour are given below:
<table><tr><th>Workers</th><th>Hours</th><th>Rate per Hour</th></tr><tr><td>Skilled labour</td><td>1,950</td><td>88</td></tr><tr><td>Semi – Skilled Labour</td><td>1,935</td><td>30</td></tr><tr><td>Unskilled Labour</td><td>3,450</td><td>44</td></tr></table>Calculate:
a) Labour Cost Variance
b) Labour Rate Variance
c) Labour Efficiency Variance</td><td>15</td></tr>
<tr><td>Q-7
CO- 2; SO-;
BL- Remember</td><td></td><td>Explain the classification of costs according to elements.</td><td>5</td></tr>
</table>

THE END